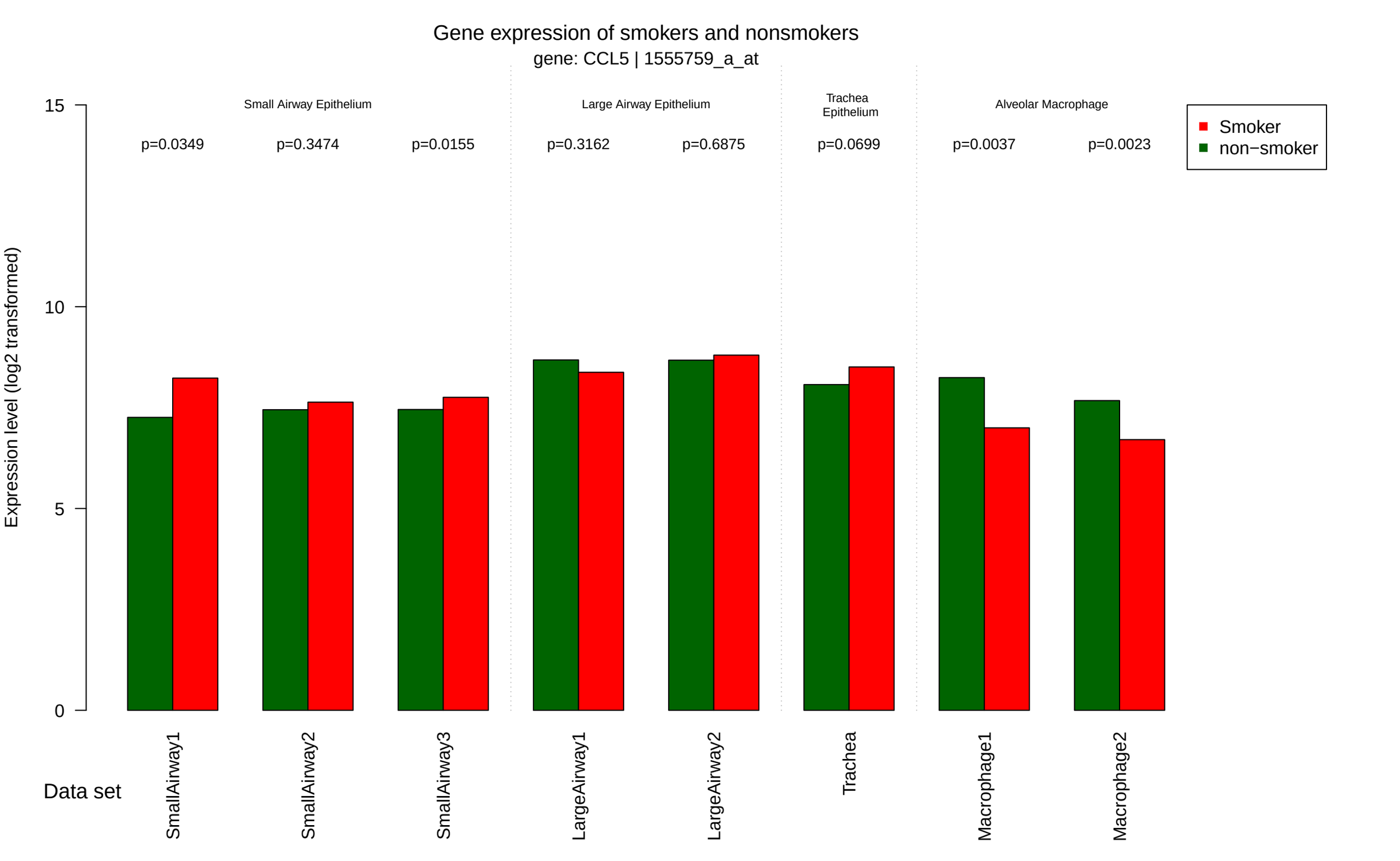
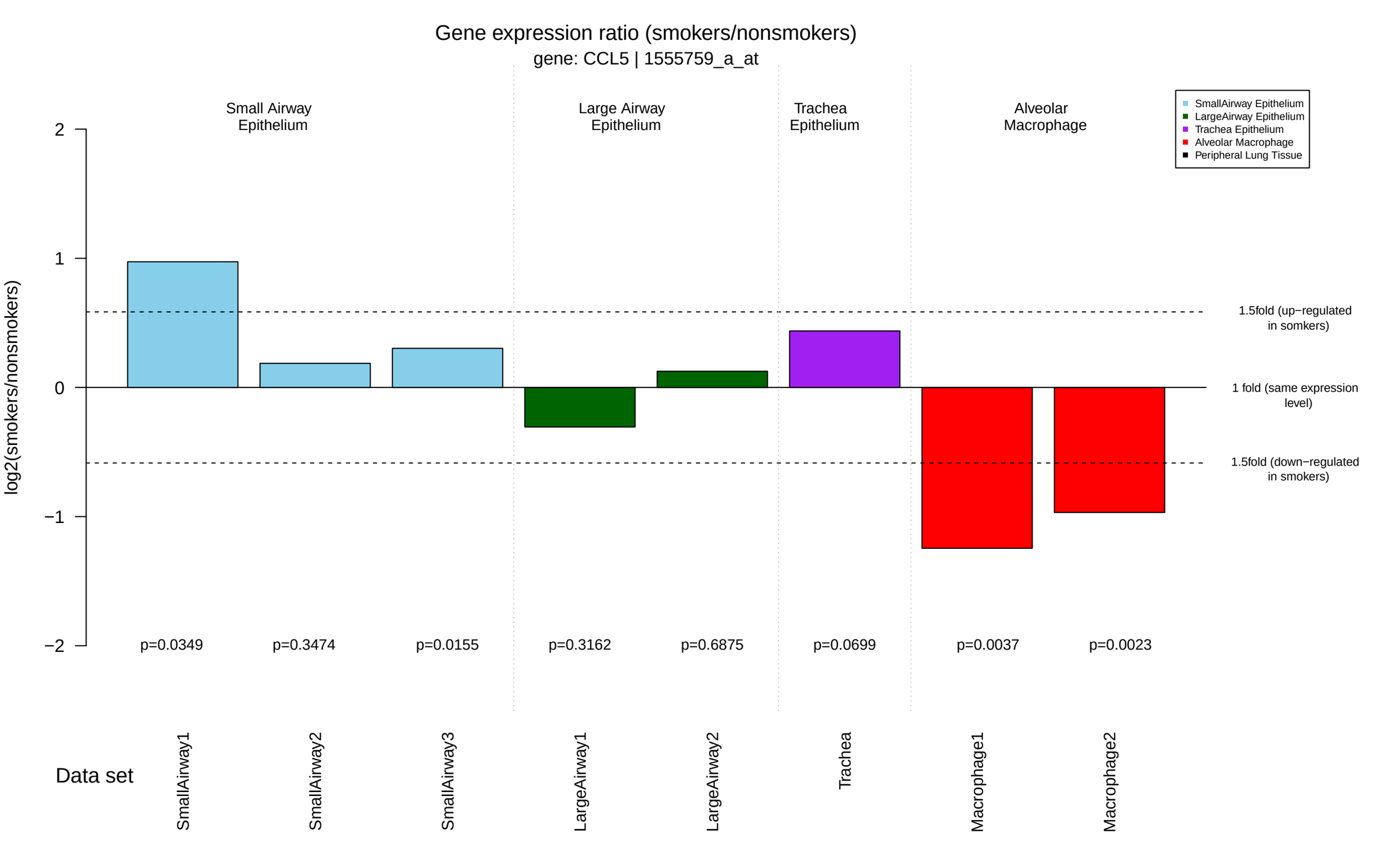
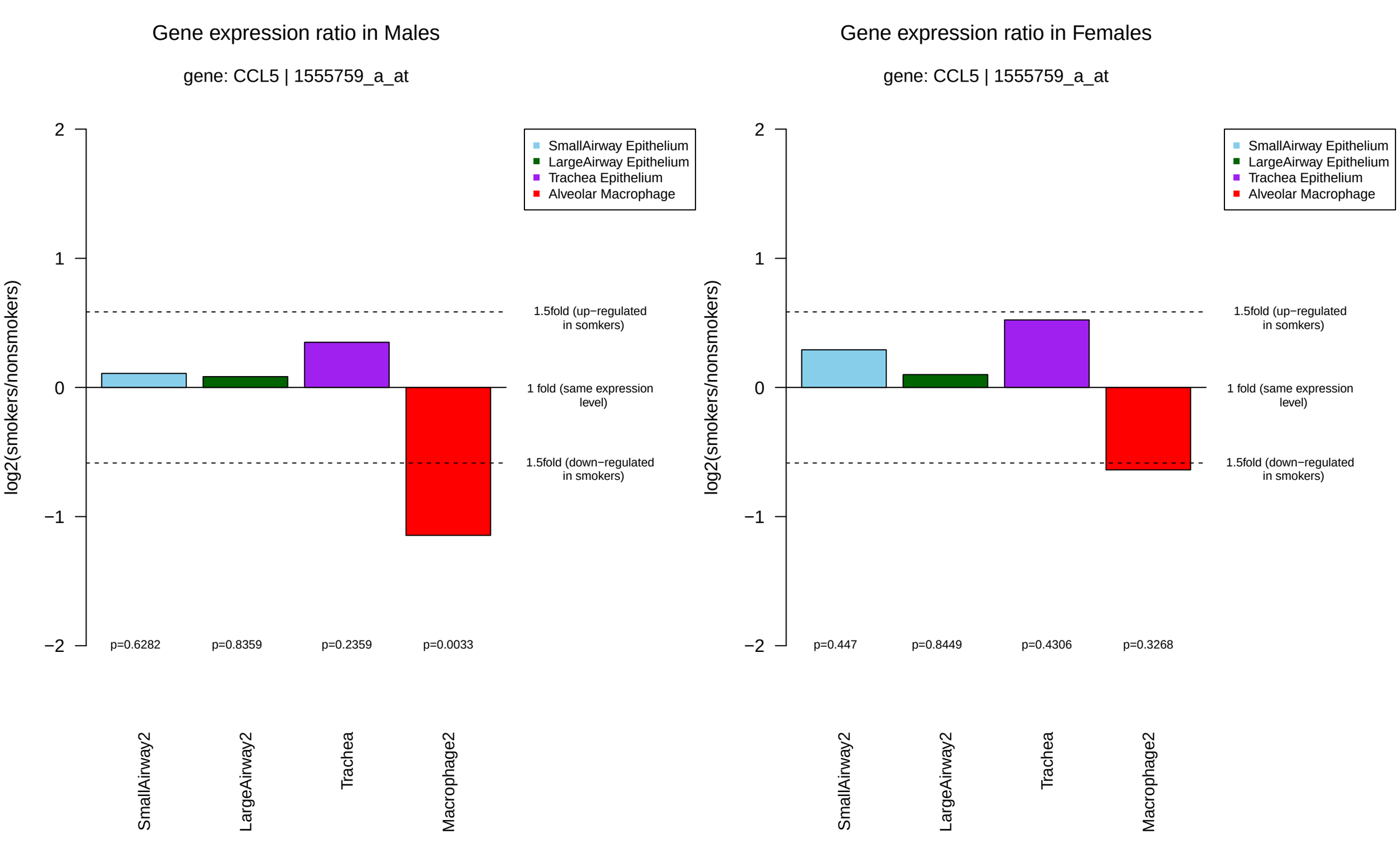
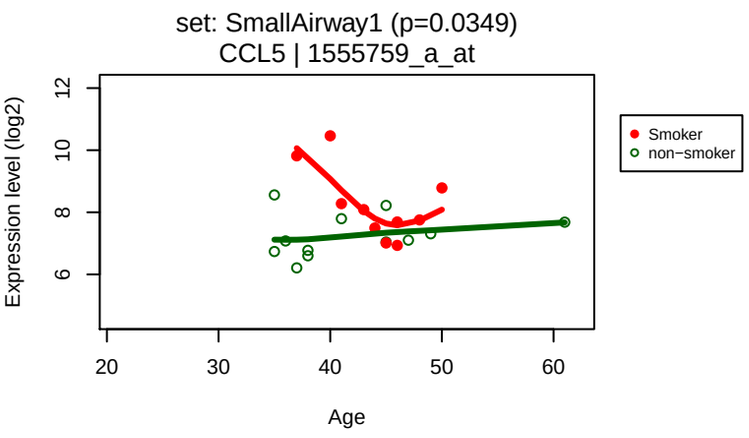
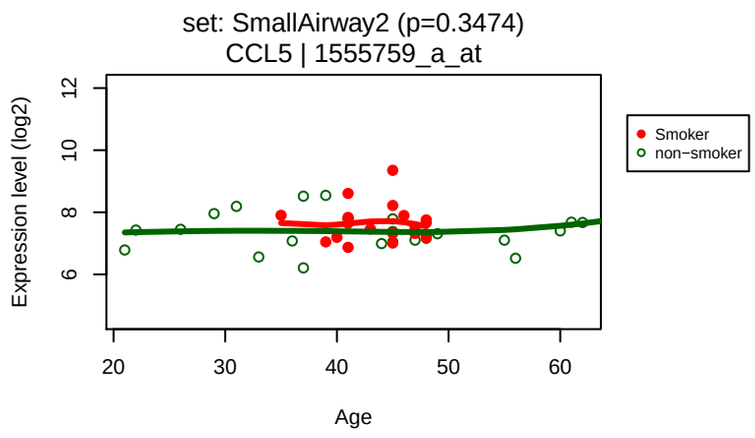
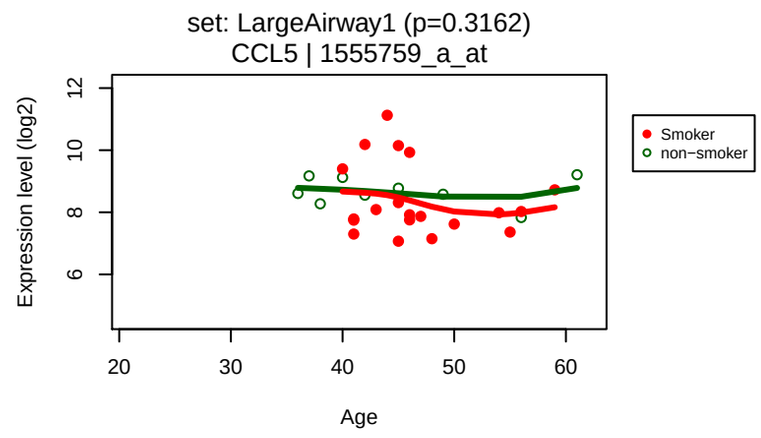

Gene expression of smokers and nonsmokers
gene: CCL5 | 1555759_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0349
p=0.3474
p=0.0155
p=0.3162
p=0.6875
p=0.0699
p=0.0037
p=0.0023
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

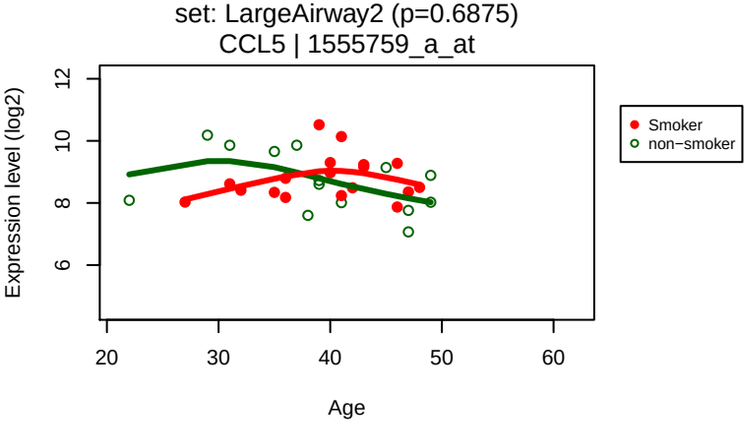

Gene expression ratio (smokers/nonsmokers)
gene: CCL5 | 1555759_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0349
p=0.3474
p=0.0155
p=0.3162
p=0.6875
p=0.0699
p=0.0037
p=0.0023
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

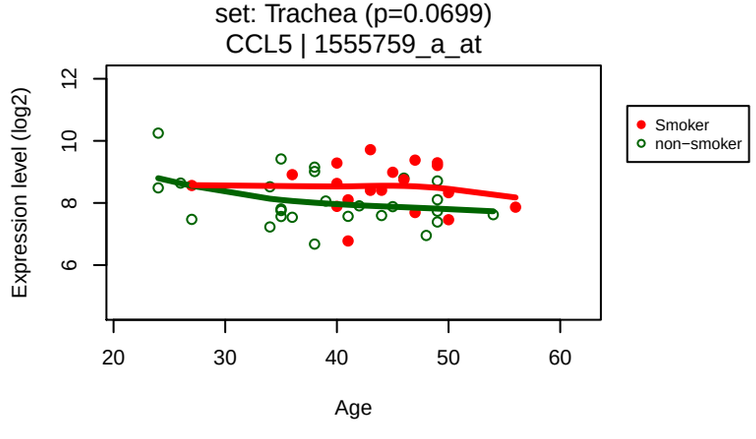

Gene expression ratio in Males
gene: CCL5 | 1555759_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6282
p=0.8359
p=0.2359
p=0.0033
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CCL5 | 1555759_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.447
p=0.8449
p=0.4306
p=0.3268
SmallAirway2
LargeAirway2
Trachea
Macrophage2

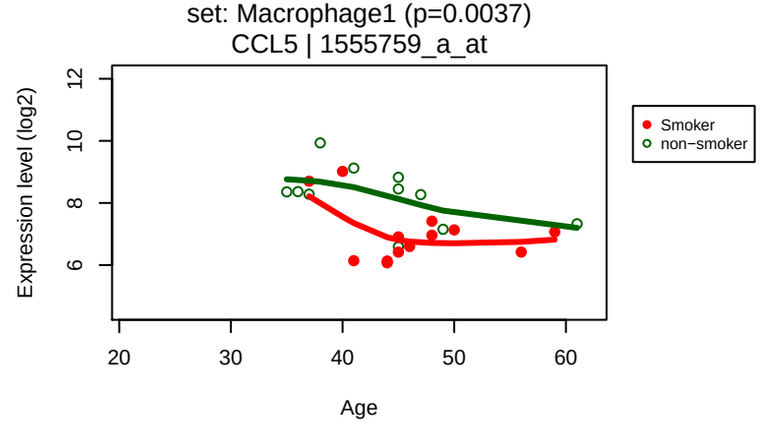

set: SmallAirway1 (p=0.0349)
CCL5 | 1555759_a_at
Expression level (log2)
Age
Smoker
non−smoker

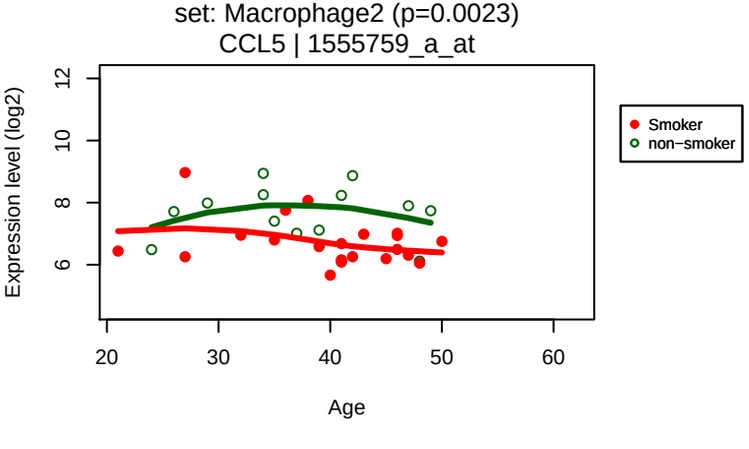

set: SmallAirway2 (p=0.3474)
CCL5 | 1555759_a_at
Expression level (log2)
Age
Smoker
non−smoker

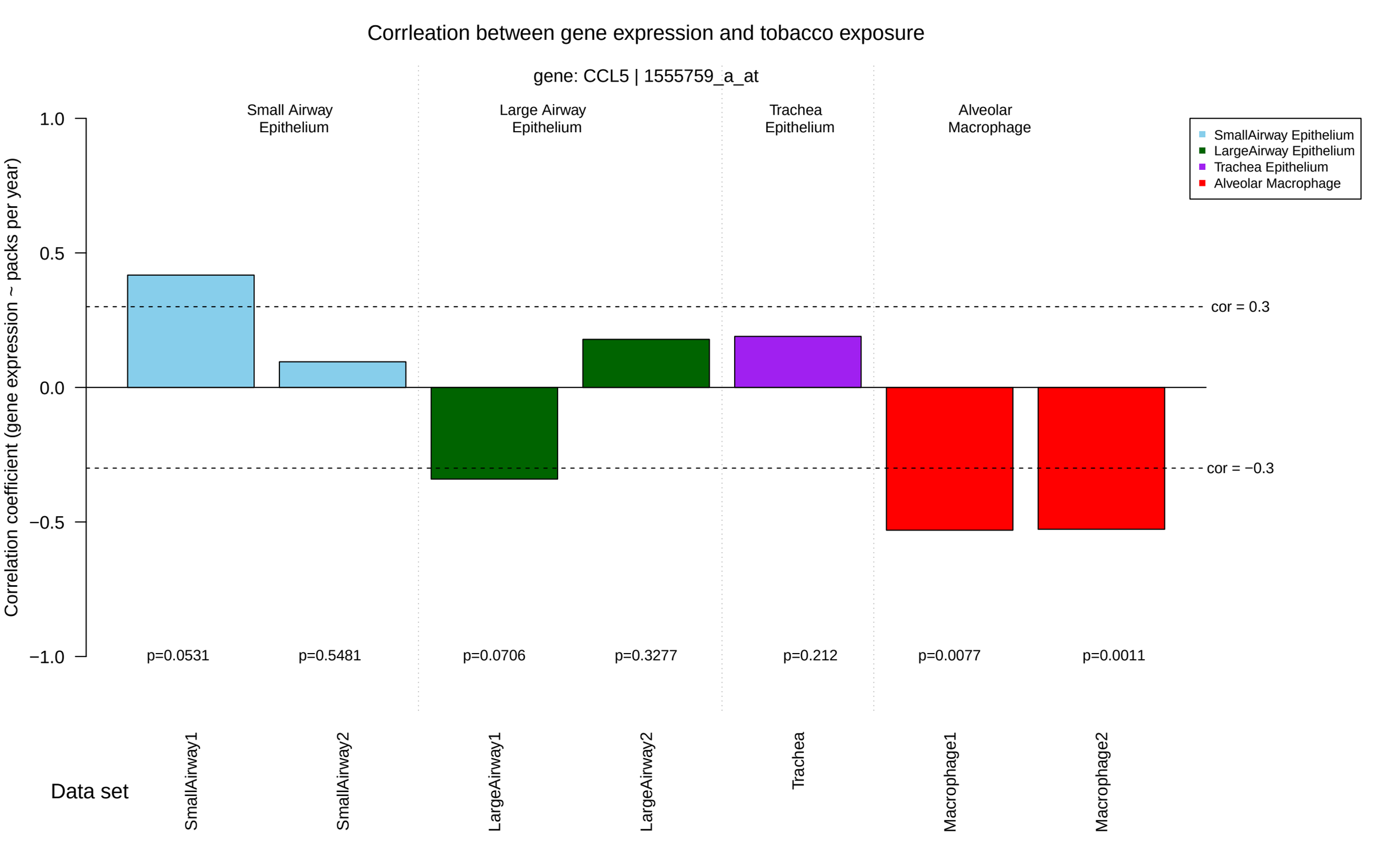

set: LargeAirway1 (p=0.3162)
CCL5 | 1555759_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6875)
CCL5 | 1555759_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0699)
CCL5 | 1555759_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0037)
CCL5 | 1555759_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0023)
CCL5 | 1555759_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CCL5 | 1555759_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0531
p=0.5481
p=0.0706
p=0.3277
p=0.212
p=0.0077
p=0.0011
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2